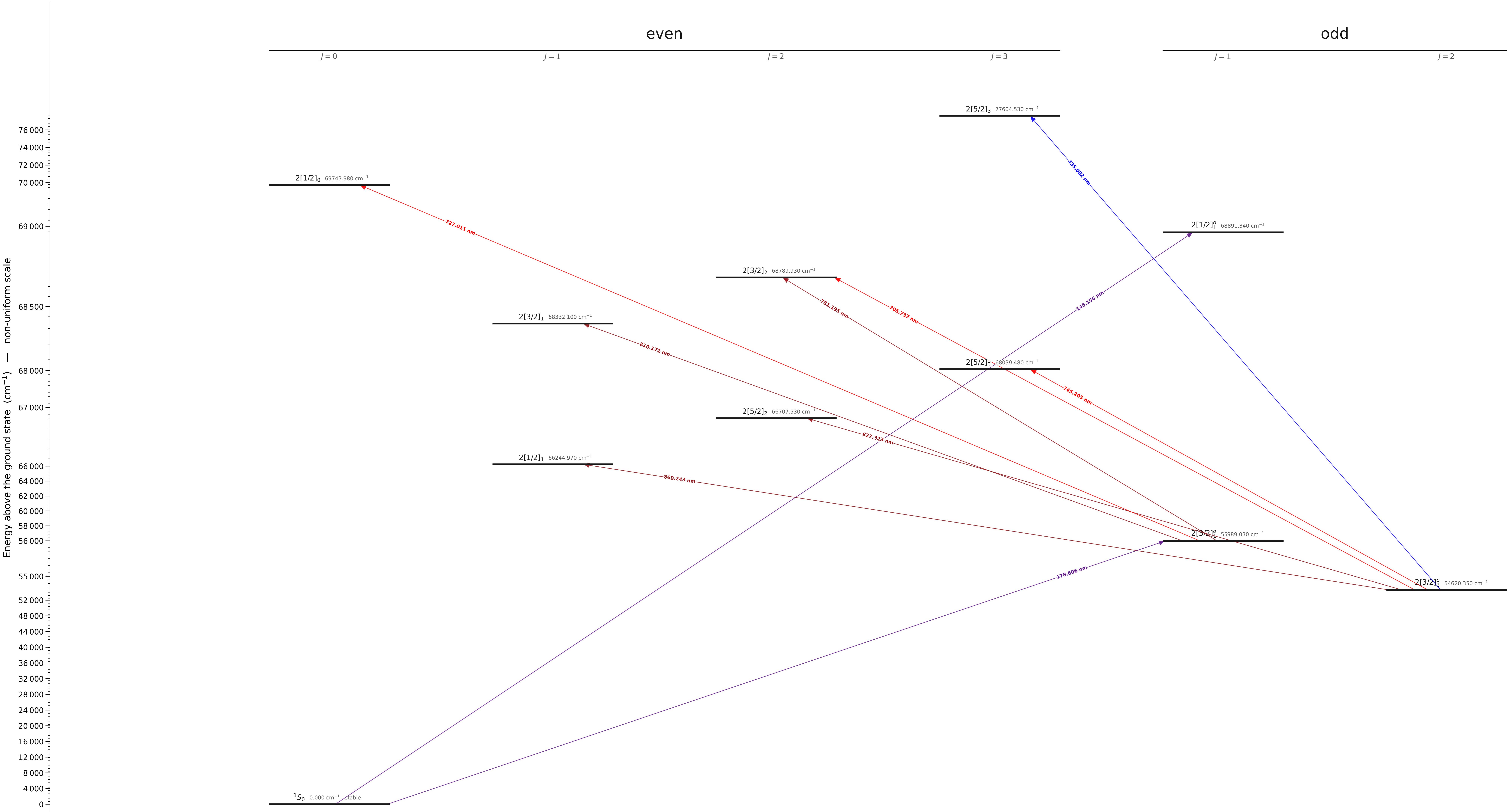
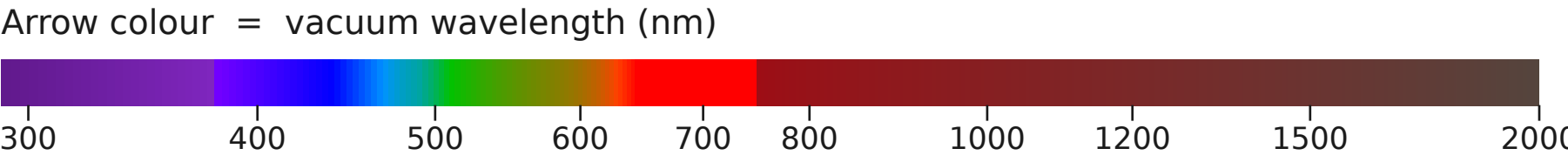


Radon

Rn (Rn I). Strongest classified transitions below 2 μm (NIST): 11 levels, 10 lines.

Rn⁺ ionisation limit 86692.50 cm⁻¹ = 10.74850 eV (115.350 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle J || er || J' \rangle |$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model or semi-empirical calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
2[3/2] ₂ ^o - 2[5/2] ₃	745.2048	744.9996	402.2954	-	-	-	-	the line of the only laser-spectroscopy study of radon: c...

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

²²¹Rn hyperfine structure (*I* = 7/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
2[3/2] ₂ ^o	-18.4(6)	2184(4)	F=3/2: +1335.81 F=5/2: +314.64 F=7/2: -568.91 F=9/2: -768.74 F=11/2: +417.30
2[5/2] ₃	-3.8(4)	2082(6)	F=1/2: +1166.25 F=3/2: +788.75 F=5/2: +258.75 F=7/2: -275.00 F=9/2: -604.25 F=11/2: -461.25 F=13/2: +481.25

Measured A, B constants; sources in the data file.